

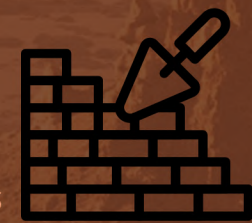
- Average solar irradiance: -590 W/m^2
- Sunlight takes approximately 13 minutes to reach Mars
- Receives 43% of Earth's solar energy
- Equatorial regions receive the most consistent solar exposure



SOLAR

- Fully subterranean habitat planning
- Thick regolith-covered protective shell
- Minimal exposed surface area
- Radiation-shielded living and sleeping spaces

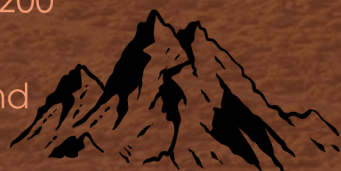
- Iron-rich basaltic regolith covers most of the Martian surface
- Locally available construction resource
- Suitable for additive manufacturing techniques
- Supports in-situ resource utilization (ISRU)



MATERIAL AVAILABILITY

- Regolith-based 3D printing
- Sulfur concrete reinforcement
- Self-sufficient construction strategy
- Reduced material transportation from Earth

- Largest canyon system in the Solar System
- Up to 7 km deep and 200 km wide
- Steep escarpments and dramatic elevation changes
- Highly suitable for sectional and terrain-responsive architecture



TOPOGRAPHY

- Multi-level indoor hydroponic farm
- Edible vegetation integrated into the habitat
- Central garden promoting food security
- Closed-loop cultivation system

- Temperature ranges from 20°C to -153°C
- Atmospheric pressure is only 0.6% of Earth's
- Extreme day-night thermal fluctuations
- Harsh and uninhabitable surface conditions

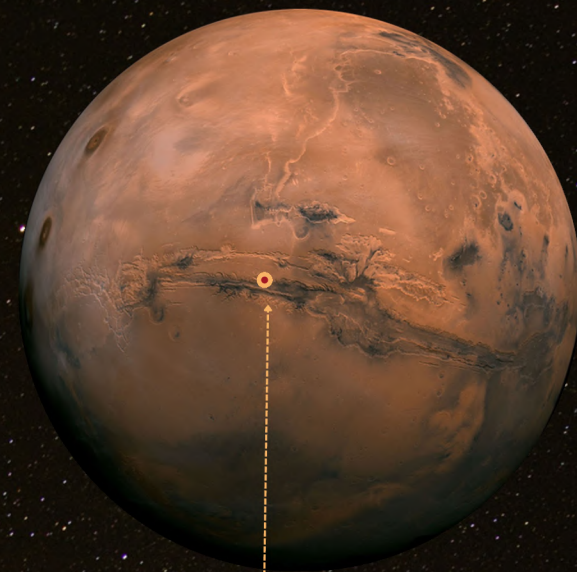


CLIMATE

- Subterranean habitat for thermal stability
- Regolith-insulated protective shell
- Pressurized indoor living environment
- Compact circular planning reduces heat loss

HER DESHER

Embedded within the cliffs of Valles Marineris, the habitat is envisioned as a self-sufficient oasis beneath the Martian surface. Organized vertically around communal and productive spaces, the design transforms a hostile landscape into a protected environment for living, cultivation, and exploration.



SITE LOCATION

The proposed habitat is located within the Melas Chasma region of Valles Marineris at approximately 11°S ; 72°W . The site is characterized by steep canyon walls, layered geological formations, and dramatic topographical conditions that support partially subterranean habitation strategies.

SWOT ANALYSIS



STRENGTHS

- Strong solar exposure near the equator
- Natural radiation protection through cliffs



OPPORTUNITIES

- Large cantilevers enabled by lower Martian gravity
- Development of self-sustaining ecosystems



WEAKNESS

- Harsh climatic conditions
- Thin atmosphere and pressure issues
- Extreme temperature fluctuations
- Difficult construction environment
- Limited natural resources



THREATS

- Dust storms
- Radiation exposure
- Psychological isolation of occupants

- Average radiation exposure: -233 mSv/year
- Approximately 40–50 times higher than Earth
- Thin atmosphere provides minimal radiation shielding
- Continuous exposure poses significant health risks



RADIATION

- Fully subterranean habitat planning
- Thick regolith-covered protective shell
- Minimal exposed surface area
- Radiation-shielded living and sleeping spaces

- Stable liquid water cannot exist on the Martian surface
- Atmospheric water vapor content: -0.03%
- Ice deposits occur beneath the surface in several regions
- Water extraction and recycling are essential for survival



WATER AVAILABILITY

- Water-efficient hydroponic farming
- Closed-loop recycling systems
- Reuse of domestic wastewater
- On-site water storage infrastructure

- Limited natural resources for food cultivation
- Absence of fertile soil and open-air agriculture
- Dependence on controlled growing environments
- Long-term self-sufficiency is essential



VEGETATION PRODUCTION

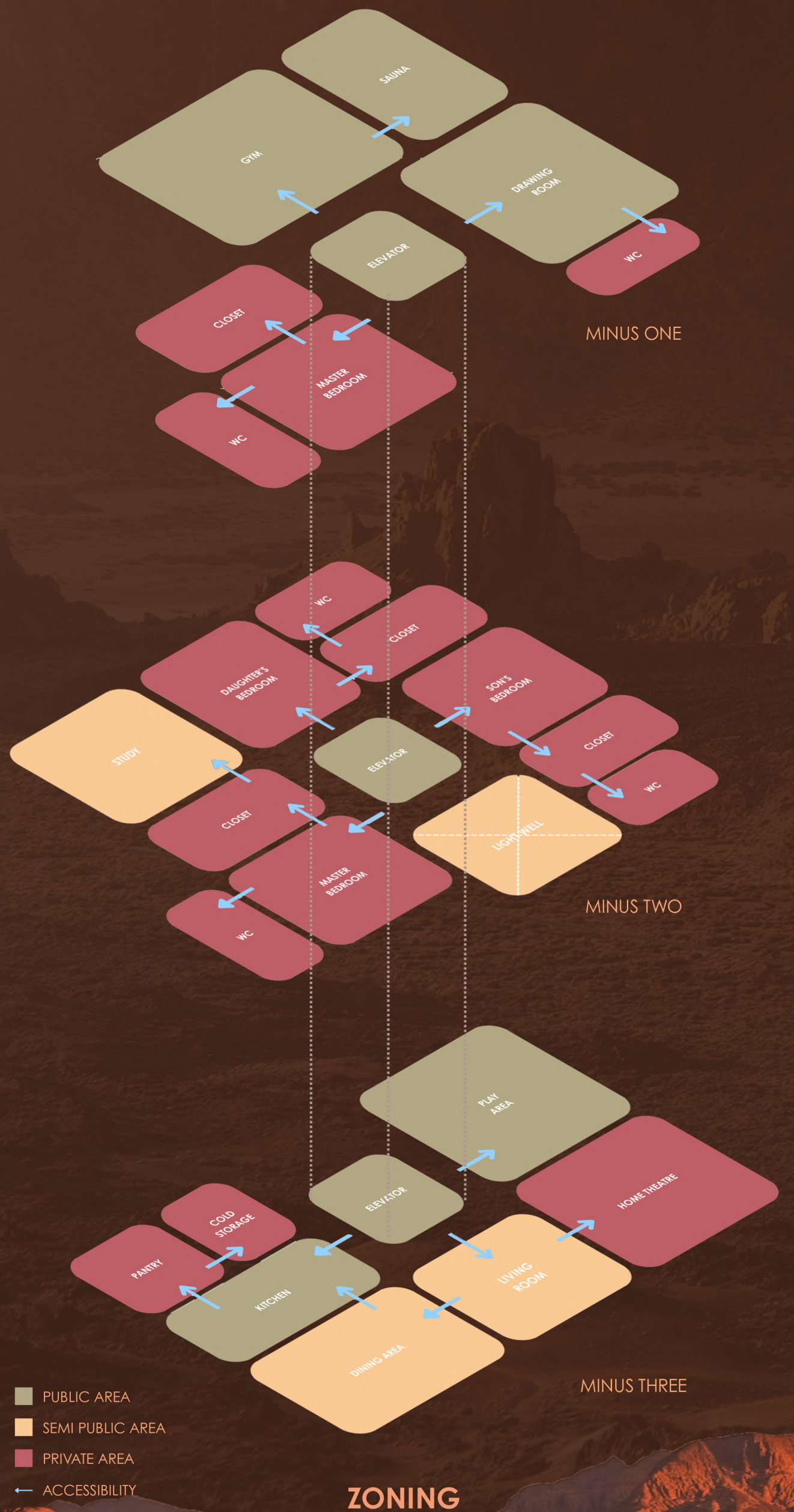
- Multi-level indoor hydroponic farm
- Edible vegetation integrated into the habitat
- Central garden promoting food security
- Closed-loop cultivation system

- Long-term isolation from Earth
- Confined underground living conditions
- Limited interaction with natural environments
- Increased risk of psychological stress

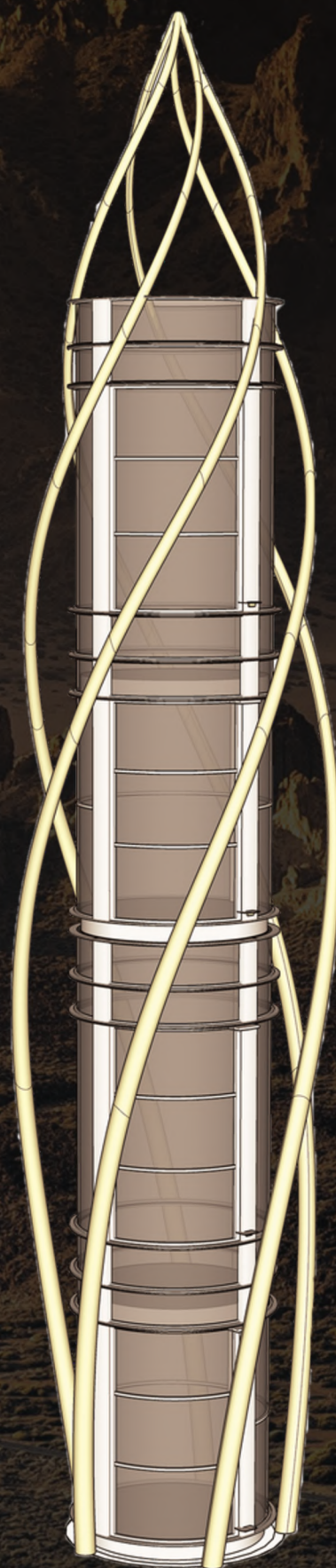


PSYCHOLOGICAL WELL-BEING

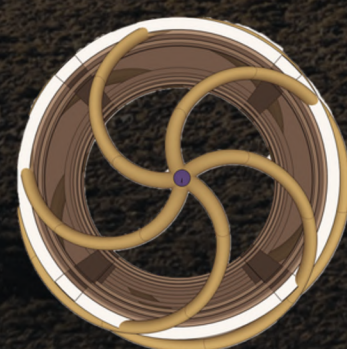
- Central biophilic garden space
- Bridges and seating integrated within vegetation
- Dedicated communal and recreational zones
- Visual connection between levels through open voids



ELEVATOR FACADE



ELEVATION



PLAN

ELEVATOR CORE

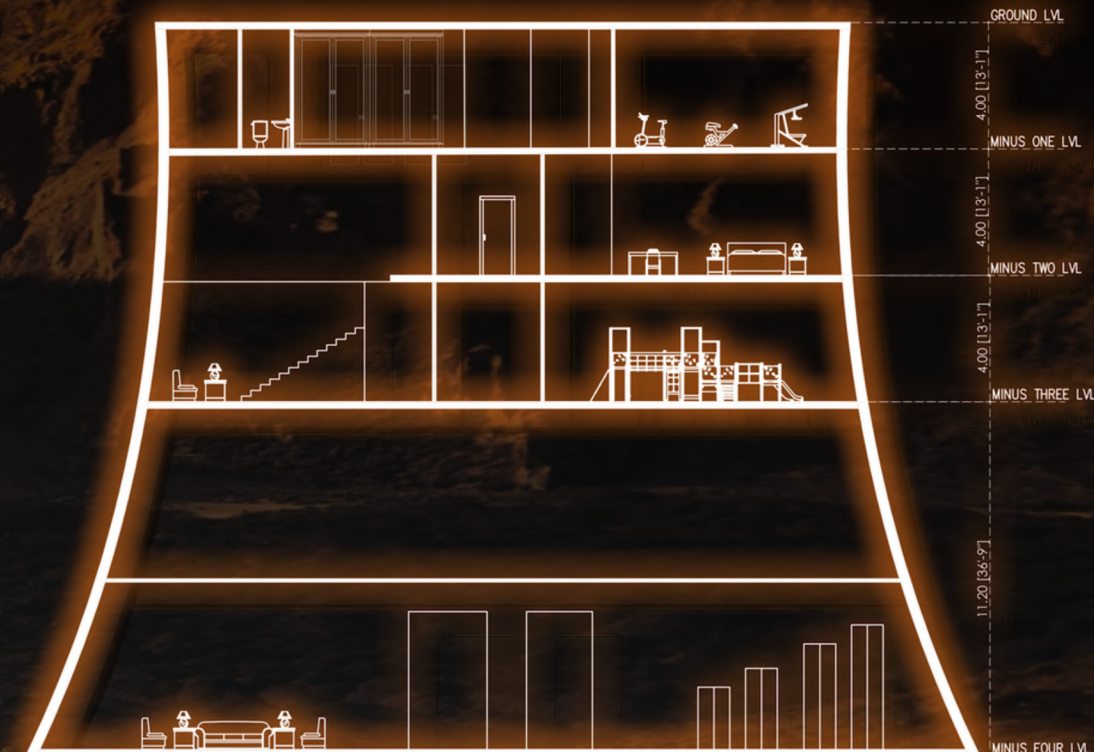
A central elevator core serves as the primary circulation spine of the habitat, connecting all underground levels. Integrated glazing around the core allows natural light from the surface to penetrate deep into the structure, improving spatial orientation and creating a visual connection between living, recreational, and productive spaces.

HABITAT SHELL

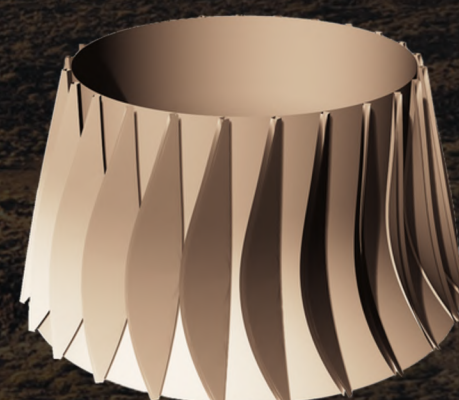
Beginning as a simple cylinder, the form evolved in response to the programmatic needs of the habitat. The lower levels were expanded to accommodate indoor farming and self-sustaining systems, resulting in a wider base. This transition was refined into a continuous protective shell, generating a structurally efficient form that integrates habitation, cultivation, and environmental protection within a unified volume.



SECTION XX'



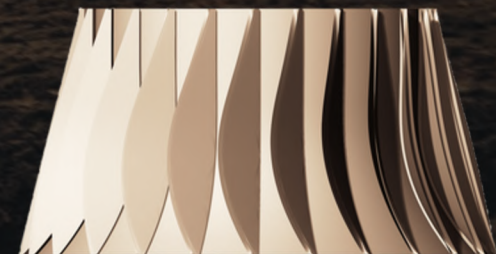
SECTION YY'



PERSPECTIVE VIEW



PLAN



ELEVATION

EXTERIOR FORM



FORM EVOLUTION



MINUS ONE



MINUS TWO



Sauna



Guest Bedroom



Study

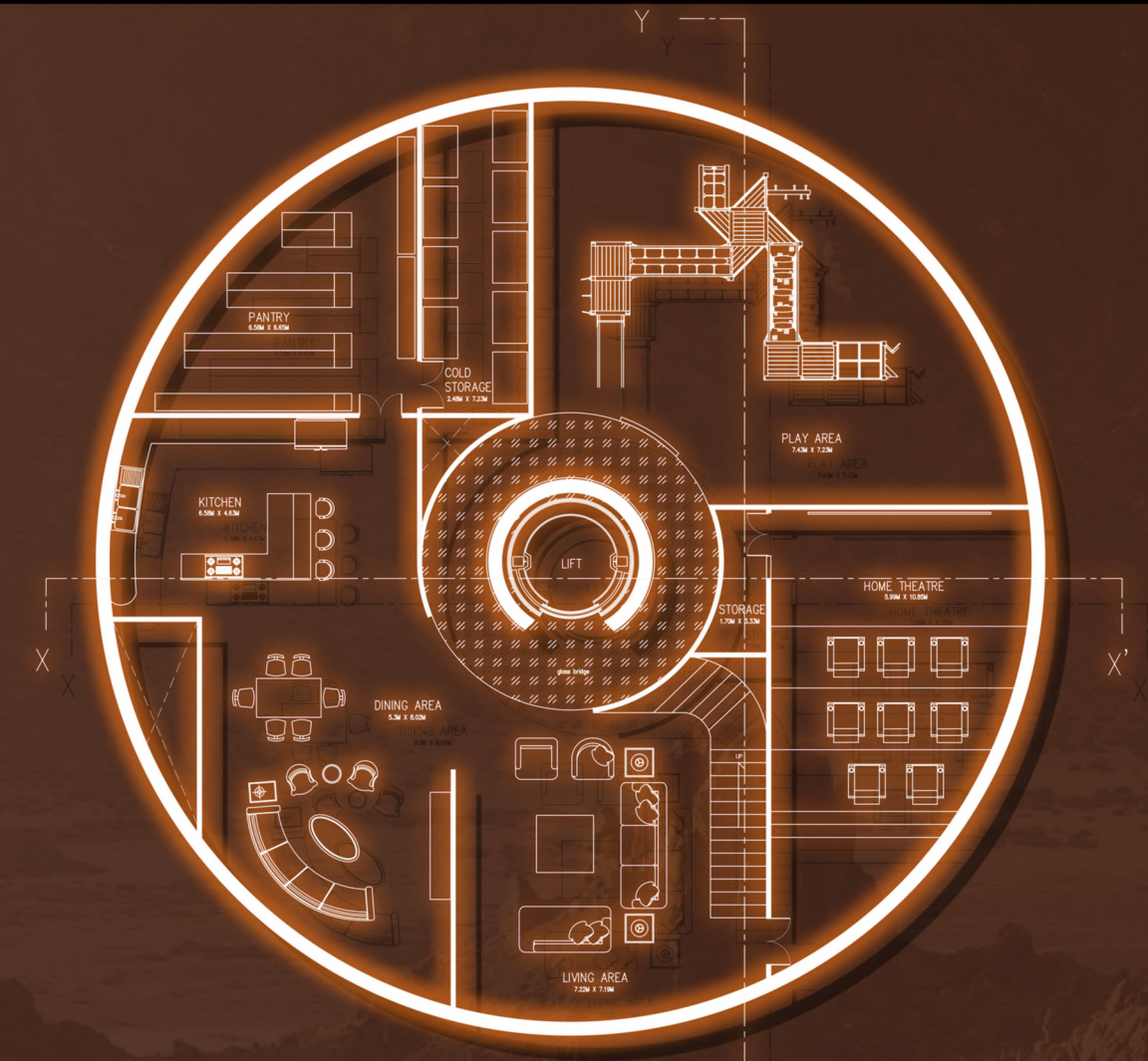
The first habitable level acts as a transitional zone between the Martian surface and the private family spaces below. A guest suite with dedicated amenities is positioned near the entrance to preserve family privacy, while a gym supports physical well-being in the low-gravity Martian environment.

Organized around a central light core, the level maintains visual connections with the lower floors through double-height spaces and glazed bridges, creating a sense of openness within the subterranean habitat.

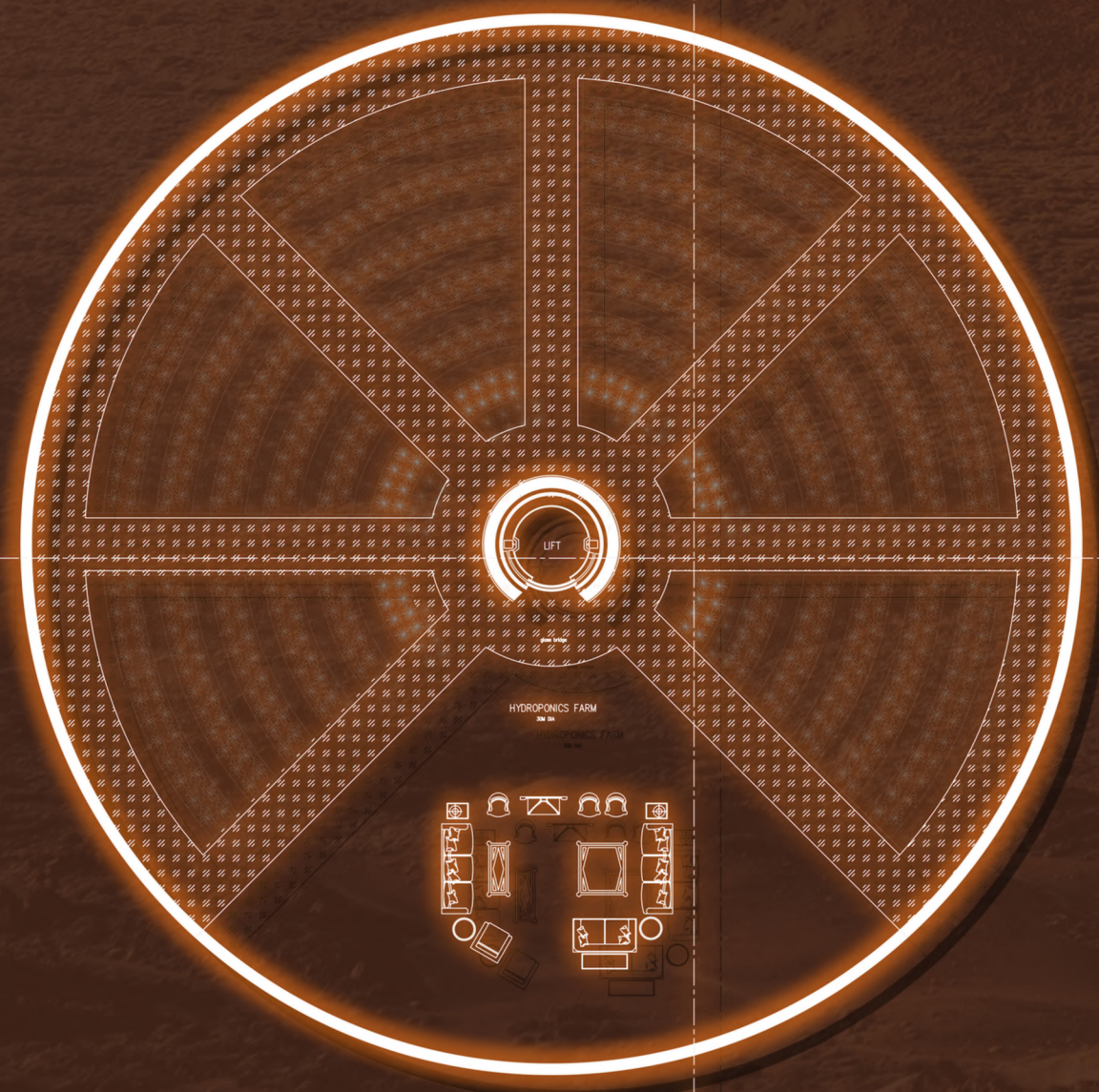
The second level serves as the private residential zone of the habitat, housing the bedrooms of all family members. Each room is provided with an attached bathroom and walk-in closet to ensure comfort and privacy. A secluded study and research space for Mr. and Mrs. Vance supports their scientific work while remaining separate from the household activities.

Glazed bridges and the central void maintain visual connections between levels, while a climbing shaft linking the children's rooms to the recreational floor encourages movement and interaction within the habitat.





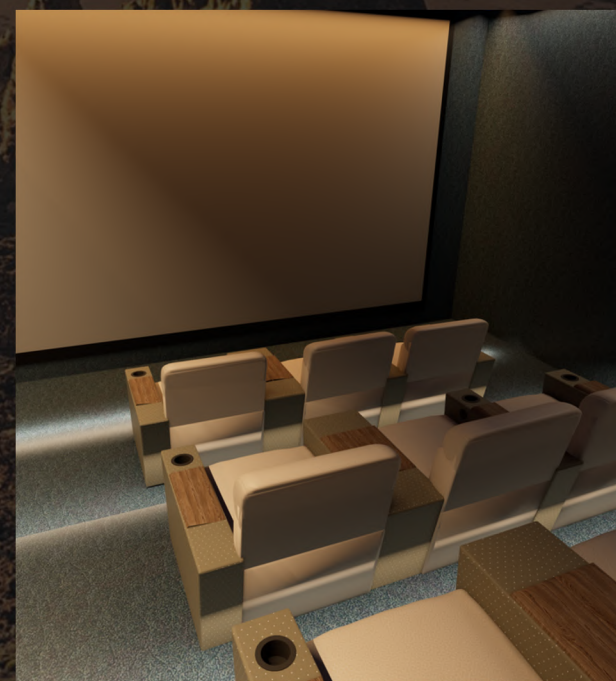
MINUS THREE



MINUS FOUR



Dining Area



Home Theatre



Hydroponic Farm

The third level functions as the primary communal hub of the habitat, bringing together recreation, dining, and everyday family activities. Connected to the private residential floor through a staircase and double-height spaces, it strengthens visual and social interaction between levels.

The floor accommodates a play area, home theatre, living room, dining space, and an open kitchen with integrated pantry storage, creating a flexible environment that supports both daily living and social gatherings. Designed as the social core of the habitat, it promotes connection, recreation, and shared experiences within the isolated Martian setting.

The lowest level functions as a hydroponic indoor garden designed to support the family's nutritional and psychological well-being. Expanded to a 15 m radius, it accommodates crops such as potatoes, tomatoes, lettuce, spinach, herbs, and strawberries using water-efficient closed-loop cultivation systems.

Integrated seating spaces and bridges allow the family to engage with nature, transforming the farm into both a productive landscape and a recreational retreat within the habitat.

